Edi Zhang

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Education

University of California, Los Angeles Los Angeles, CA

BS, Computer Science and Applied Mathematics - 3.81/4.00 GPA

- **Math/EECS Coursework**: Data Structures, Algorithms, Systems and Signals, Convex Optimization (Graduate), Software Construction, Computer Organization, Real Analysis I/II, Abstract Algebra, Discrete Math
- **Machine Learning Coursework:** Neural Networks/Deep Learning I/II (Graduate), Applied Numerical Computing, Abstract Linear Algebra (Honors), Computational Imaging (Graduate), Probability, Statistics;
- **Activities and Societies:** Club Swimming National Finalist, ACM ICPC, Nova Tech for Good, The Bruin Group Consulting, Bruin Sport Analytics, Campus Barber, Awaken Acapella, Chess Club

Technical Skills

Languages: Python, C/C++, JavaScript/TypeScript, Java, GraphQL, SQL, HTML/CSS, Julia, MATLAB
Frameworks/Libraries/Tools: React, Next.js, PyTorch, NumPy, Pandas, Matplotlib, MongoDB, Git, Figma

Experience

Software Engineering Intern June 2026 – September 2026

Atlassian — Loom *San Francisco, CA*

- Architecting and implementing an end-to-end offline recording feature for Loom Desktop app using TypeScript, enabling uninterrupted video capture for over 14M+ active users during critical network disruptions/outages.
- Engineered a serverless recording architecture in TypeScript, enabling asynchronous video capture startup to eliminate server wait times and ensure instant recording initialization.
- Implemented local key caching and robust Electron recovery queues, guaranteeing seamless video data preservation and automatic background synchronization upon network reconnection.

Software Engineering Intern June 2025 – August 2025

Capital One *Richmond, VA*

- Developing metric summaries to visualize compliance vulnerabilities across internal systems using ReactJs/Next.js
- Tested and deployed app to a central data platform with 4600+ weekly users, improving developer efficiency by 25%
- Transformed manual data-pulling into automated data workflows by creating DAGs using Apache Airflow and PySpark for scheduled batch processing of vulnerability reports, allowing for consistent summary updates

Researcher June 2024 – March 2025

Cleveland Clinic Lerner Research Institute *Cleveland, OH*

- Researched in the Advanced Musculoskeletal Imaging Lab on mathematical modeling of tibiofemoral joint using Generalized Procrustes and Principal Component statistical analyses to analyze directions of bone shape variance
- Constructed topological surfaces by processing 275 MRI masks using Marching Cubes and Laplacian Smoothing
- Presented abstract at 2025 Orthopaedic Research Society Annual Conference to leading scientists and researchers

Projects

EMG Classification | *Python, PyTorch*

- Tested EMG Encoder to measure keystroke classification accuracy using architectures including CNNs (75.41%), LSTMs (77.52%), GRUs (77.05%), CNN + LSTM (79.47%), and CNN + GRU (80.16%) on emg2qwerty datatset
- Achieved 80.55% accuracy by fine-tuning using CNN + GRU + Gaussian Noise and analyzed other combinations of fine-tuning methods including Frequency masks, Downsampling, and Random Cropping to optimize performance

PyTorch Implementation | *Python, NumPy*

- Built FFNNs and CNNs and implemented functions and parameters such as backprop, batch-norm, k-nearest neighbors, k-means, softmax classifier, Adam/RMSProp/SGD optimizers, momentum, dropout using NumPy
- Optimized and tested these Neural Networks on CIFAR-10 to achieve 75% validation accuracy on image classication

Honors and Awards

USA Coding Olympiad (USACO) Gold Division

3x American Invitational Math Exam (AIME) Qualifier